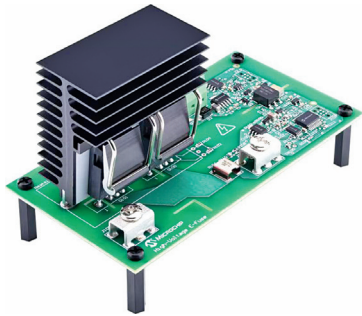
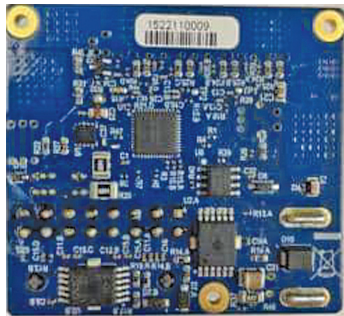


Interesting Reference Designs

Title of Reference Design	High-Voltage Auxiliary E-Fuse	Power Distribution Box With E-Fuse
Image		
Key Attractions	The reference design is tailored for hybrid electric vehicles (HEVs) and electric vehicles (EVs). It features a high-voltage solid-state relay with overcurrent protection, ensuring optimal performance and efficiency.	The electronic fuse (e-fuse) box reference design is a practical solution for power distribution systems. Despite its compact footprint, the board houses ten output channels, all operated by Renesas Intelligent Power Devices (IPDs).
Highlights	The high-voltage auxiliary e-fuse reference design from Microchip is launched specifically for HEVs and EVs. Leveraging Microchip's 700V and 1200V silicon carbide (SiC) technology, along with other unique innovations, the demonstrator provides an all-encompassing total system solution (TSS). The design integrates a time-current characteristic (TCC) curve, making it suitable for non-automotive applications, especially DC solid-state circuit breakers. It boasts a high-voltage solid-state relay with overcurrent protection for enhanced efficiency. This consolidated approach eliminates the need for individual components, promises reliability, and accelerates development. The design also offers a configurable current limit profile, a LIN communication interface for effortless configurability, and the option for high-side or low-side drive configuration. The robust e-fuse can endure short circuits up to 10μs, functions efficiently within voltage ranges of 9V-16V and 200V-900V, and operates reliably between -40°C and 85°C. Built with AEC-qualified components, it supports 400V and 800V bus voltages, has current ratings up to 30A, and switches at frequencies as high as 20kHz.	The Y-IPD-EFUSE-PDB-01, a reference design from Renesas, offers practical solutions for power distribution systems. The e-fuse functionality capitalises on the feedback from these IPDs, with the RL78 microcontroller unit (MCU) software overseeing its operation. The board stands out for its adaptability, with adjustable fuse characteristics suitable for various cables and settings. It comprises ten switched outputs with varying capacities, including two channels at 30A, four channels at 10A, and four channels at 3A, all fitting within its compact dimensions measuring 55mm by 55mm by 25mm. Additional features include a Renesas ISL78301 LDO for a 5V power supply, an integrated negative temperature coefficient (NTC) for temperature accuracy, and functionality in -40°C to +85°C environments. Communication is streamlined via a controller area network (CAN) interface, complemented by sturdy connectors and a user-friendly GUI. From a top view, the board displays soldered components like IPDs of varying capacities, a linear regulator, and multiple connectors. The underside houses two 3A IPDs, a 16-bit MCU, LIN and CAN transceivers, and seven user-friendly LEDs indicating operational modes.
Applications	The reference design is optimal for both on-board and off-board chargers and energy storage systems (ESS).	The reference design is suitable for a quick charging solution without overburdening residential and public electrical systems.
OEM Brand	Microchip	Renesas
URL	https://www.electronicsforu.com/electronics-projects/high-voltage-auxiliary-e-fuse-reference-design	https://www.electronicsforu.com/electronics-projects/reference-design-for-power-distribution-box-with-e-fuse